

FlowSentinel AI: Real-Time Demo Overview

TO: Engineering Leadership & Review Committee

DATE: August 25, 2026

FROM: CTech Data Engineers

SUBJECT: RT Demo Proposal & Architecture

1. The Core Problem Statement

Why Data Engineering On-Call Needs Transformation:

- **High MTTR (2 to 6 hours):** Debugging Airflow/BigQuery logs manually consumes up to 40% of engineer capacity.
- **Execution Fatigue:** Off-hours manual SQL hotfixes and backfill script execution risk human error and quota crashes.
- **Data Poisoning:** Late vendor CSV/SFTP feeds propagate downstream, corrupting executive analytics dashboards.
- **Backfill Uncertainty:** Calculating historical partition impact and BigQuery compute costs is based on manual guesswork.

2. Real-Time (RT) Demo Breakdown: What the Demo Consists Of

Act 1: Live Pipeline Failure & Autonomous Triage

 3 Mins

Inject a synthetic schema-drift failure into a running Cloud Composer DAG feeding BigQuery. Cloud Logging routes failure to Pub/Sub. **Gemini 2.0 Triage Agent** parses 500 lines of logs, creates a structured **Jira/GitHub ticket** with exact line of fault, and updates Dashboard status to  **Action Required**.

Act 2: Autonomous Code Patch & 1-Click Approval

 3 Mins

Code Repair Agent drafts a GitHub Pull Request with schema fix. Engineer receives a Slack notification, views PR diff on the **React/Cloud Run Ops Dashboard**, and taps **[Approve & Auto-Deploy]** to merge & redeploy instantly without manual coding.

Act 3: Code-Change Driven Precision Backfill

 3 Mins

Backfill Agent parses Dataform/Airflow DAG lineage graph, identifies 7 days of affected historical partitions, estimates BigQuery compute cost (\$1.20), and presents a 1-click execution card on the Dashboard to run sequential backfill jobs safely.

Act 4: Upstream Source Feed Sentinel & SLA Guardrail

 3 Mins

Simulate delayed 3rd-party vendor CSV in GCS. **Source Feed Sentinel** detects SLA breach, auto-pauses downstream DAGs to prevent data corruption, dispatches vendor alert, and auto-resumes processing immediately once the file lands.

3. Expected Value & Competitive Advantage

Operational Metric	Standard Industry Demos	FlowSentinel AI Framework
MTTR (Mean Time to Resolution)	2 – 6 Hours	< 2 Minutes
Human Labor Requirement	Manual log analysis & SQL scripting	1-Click Review & Governance
Platform Portability	Custom script lock-in	Terraform-ready for any client GCP setup

